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Autonomous Institute, Affiliated to VTU

Department of Artificial Intelligence & Machine Learning

AML23702

Advance Natural Language Processing

AML23702_Syllabus.pdf

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Module-5
Word Senses and WordNet

Word Senses and WordNet: Word Senses, Relation between Senses, WordNet: A Database of Lexical relations, Word Sense disambiguation, Alternate WSD algorithms and Tasks, Word Sense Induction

Textbook: M5_Word Senses and WordNet.pdf

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Pre-session Reading

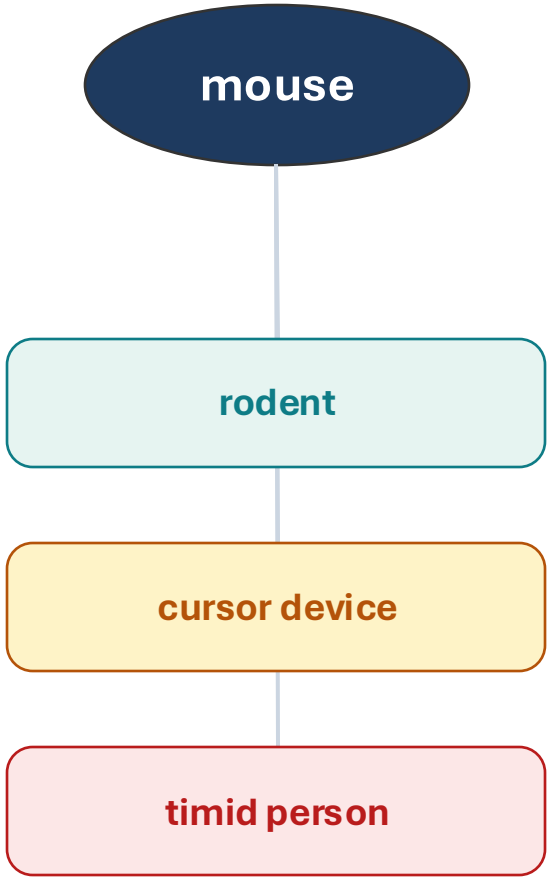
- A reading on WordNet as a lexical database, how synsets are organized, and how senses are linked through semantic relations - https://wordnet.princeton.edu/homepage
- Short material on Word Sense Disambiguation, including dictionary-based, supervised, and unsupervised approaches - https://www.geeksforgeeks.org/machine-learning/word-sense-disambiguation-in-natural-language-processing/
- Word senses, ambiguity, and how a single word can have multiple meanings depending on context - https://web.stanford.edu/~jurafsky/slp3/old_sep21/18.pdf

APPENDIX I · SPEECH AND LANGUAGE PROCESSING

Word Senses and WordNet

Polysemy · sense relations · WordNet · word sense disambiguation · Lesk · sense induction

Jurafsky & Martin, draft of August 24, 2025



one word · many senses

What this chapter covers

Roadmap

1

Word senses

Polysemy, glosses, how many senses a word has

2

Sense relations

Synonymy, antonymy, hypernymy, meronymy, metonymy

3

WordNet

Synsets, supersenses, relation inventory, IS-A chains

4

Disambiguation

Nearest-neighbour WSD, features, Lesk, WiC

Why it matters: question answering (“bat care”), machine translation (murciélago vs. bate), and evaluating how language models represent meaning.

01

Word Senses

A sense is a discrete representation of one aspect of a word's meaning.

Words are ambiguous

The same word form can be used to mean different things. Words with many senses are polysemous (Greek poly- “many” + sema “sign”).

mouse

mouse[1] a small rodent

mouse[2] a device that controls a cursor

bank

bank[1] a financial institution

bank[2] a sloping mound of land

lose

lose[1] to misplace an object

lose[2] to suffer someone's death

“To lose one parent may be regarded as a misfortune; to lose both looks like carelessness.”
Oscar Wilde, The Importance of Being Earnest

Senses are clear in context

Lexicographic tradition marks each sense with a superscript. In running text the intended sense is usually obvious.

mouse[1]	*… a mouse controlling a computer system in 1968.*
mouse[2]	*… a quiet animal like a mouse …*
bank[1]	*… a bank can hold the investments in a custodial account …*
bank[2]	*… as agriculture burgeons on the east bank, the river …*

Embeddings (word2vec, GloVe) define a word by its co-occurrences — but they do not tell us how to define a sense. Contextual embeddings (BERT) do, and they sit at the heart of modern WSD.

Defining a sense: glosses

Dictionaries and thesauruses give a textual definition — a gloss — for each sense, often with example sentences.

bank[1]

financial institution that accepts deposits and channels the money into lending activities

bank[2]

sloping land (especially the slope beside a body of water)

Not a formal representation

- Glosses are written for people, not machines
- But a gloss is a sentence — so it can be turned into a sentence embedding
- Example sentences help build the sense representation too

Second route: relations

- Define a sense by how it relates to other senses
- right / left stand in opposition; red IS-A colour; blood IS-A liquid
- These relations are listed explicitly in WordNet

The circularity of dictionary definitions

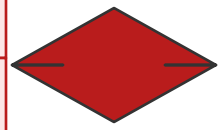
American Heritage Dictionary

Definitions lean on each other. Useful for a human reader who already grasps the other terms — a problem for a machine.

Glosses are still useful: a gloss is a sentence, and sentences have embeddings.

right adj. located nearer the right hand …

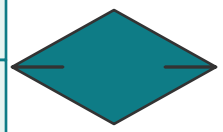
left adj. located nearer to this side of the body than the right



right and left each define the other

red n. the colour of blood or a ruby

blood n. the red liquid that circulates in the heart …



red and blood each define the other

How many senses does a word have?

Dictionaries give discrete lists; embeddings give a continuous space. So we need criteria for splitting uses into senses.

Four tests for a distinct sense

1. Independent truth conditions
2. Different syntactic behaviour
3. Independent sense relations
4. Antagonistic meanings

Three senses of serve (WSJ corpus)

serve[1] — provide food
They rarely serve red meat, preferring to prepare seafood.

serve[2] — hold office — serve as NP
He served as U.S. ambassador to Norway in 1976 and 1977.

serve[3] — spend a sentence
He might have served his time, come out and led an upstanding life.

The zeugma test

Conjoin two uses of a word in one sentence. If the result sounds odd, the two uses belong to different senses.

✓ *Which of those flights serve breakfast?*

✓ *Does Air France serve Philadelphia?*



? *Does Air France serve breakfast and Philadelphia?*

Zeugma
The oddness shows there is no single sense of serve that covers both breakfast and Philadelphia.

Granularity
Dictionaries prefer fine-grained senses for learners. For computation we usually cluster them into coarser groups.

02

Relations Between Senses

Synonymy, antonymy, taxonomy, meronymy and structured polysemy.

The map of sense relations

Synonymy

couch / sofa

near-identical senses

Antonymy

long / short

opposite on one dimension

Hyponymy

car IS-A vehicle

more specific subclass

Hypernymy

vehicle ⊃ car

superordinate class

Meronymy

wheel part-of car

part–whole relation

Metonymy

bank: building ↔ org.

structured polysemy

All of these hold between senses, not between word forms — and all of them are recorded in WordNet.

Synonymy holds between senses

Classic synonym pairs

couch ≈ sofa

vomit ≈ throw up

filbert ≈ hazelnut

car ≈ automobile

big vs. large — synonymous only in some senses

✓ *How big is that plane?*

✓ *Would I be flying on a large or small plane?*

✓ *Miss Nelson became a kind of big sister to Benjamin.*

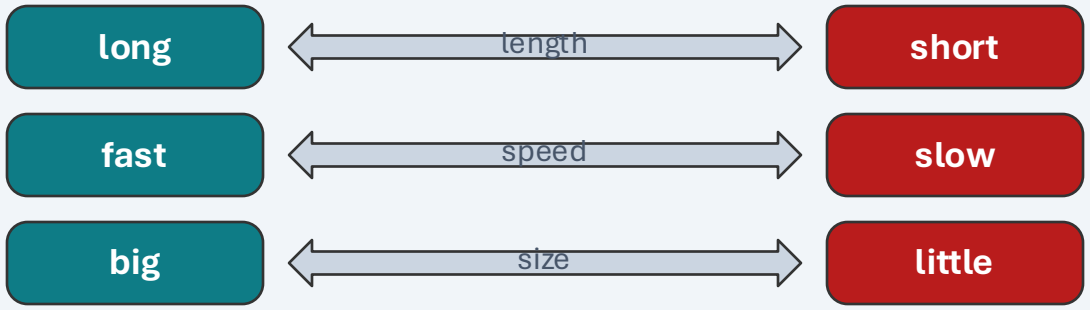
? *Miss Nelson became a kind of large sister to Benjamin.*

big has a sense meaning “older, grown up” that large lacks. So some senses of big and large are synonymous and others are not — which is why synonymy is a relation between senses.

Antonymy: opposite on one dimension

Antonyms differ completely in one aspect — position on a scale or direction of change — but share almost everything else.

Scalar opposites — ends of one scale



Reversives — opposite direction

rise ⇄ fall

up ⇄ down

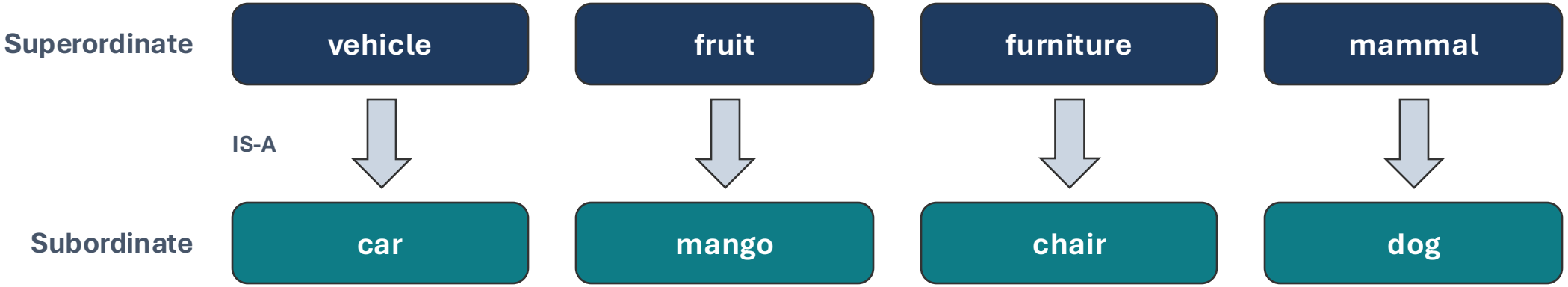
in ⇄ out

Why this is hard for embeddings

Because antonyms share nearly all their meaning, one of the nearest neighbours of up in word2vec space is often down. If a user asks an agent to “turn up the music”, confusing the two is costly. A thesaurus that records antonymy can fix what the embedding cannot.

Taxonomy: the IS-A hierarchy

A sense is a hyponym of another if it is more specific. The converse is the hypernym, also called the superordinate.



Extensional

The class denoted by the superordinate includes every member of the hyponym class.

Entailment

A is a hyponym of B if $\forall x\ A(x) \Rightarrow B(x)$ — being an A entails being a B.

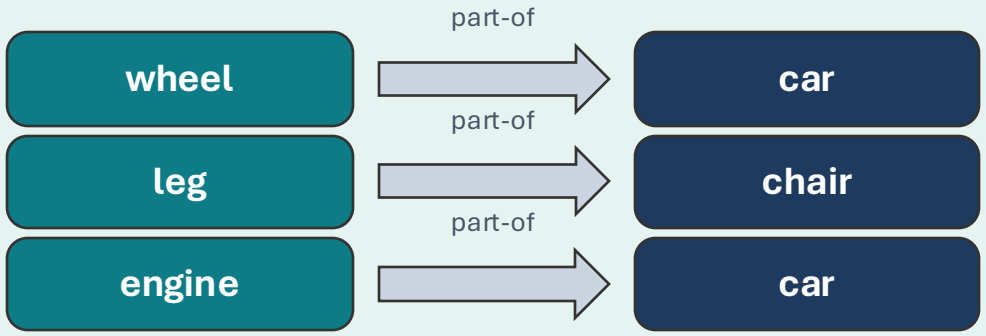
Transitive

If $A \subset B$ and $B \subset C$, then $A \subset C$. Useful for entailment and question answering.

Knowing that leukemia IS-A cancer is exactly what lets a QA system answer questions about leukemia.

Part–whole and structured polysemy

Meronymy — the part-whole relation



wheel is a meronym of car; car is a holonym of wheel.

Metonymy — one aspect stands for another

BUILDING ↔ ORGANIZATION		*school, hospital, bank*
AUTHOR ↔ WORKS		*I really love Jane Austen*
FRUIT TREE ↔ FRUIT		*I ate a preserved plum*

Structured polysemy in action

“The bank is on the corner of Nassau and Witherspoon.”

This bank[4] means the building belonging to a financial institution. The organisation ↔ building pairing recurs across school, university and hospital — a systematic relation, not an accident. That is why we call it structured polysemy, and this subtype metonymy.

03

WordNet

A large lexical database of senses and the relations between them.

WordNet 3.0 at a glance

Three separate databases — nouns, verbs, and adjectives+adverbs. Closed-class words are not included.

117,798

Nouns

1.23 senses avg.

11,529

Verbs

2.16 senses avg.

22,479

Adjectives

4,481

Adverbs

Lemma

A word form, annotated with a set of senses

Synset

The set of near-synonyms expressing one concept — WordNet's primitive unit

Gloss

A dictionary-style definition, a property of the synset

Supersense

A coarse lexicographic category (26 for nouns)

Because they share a gloss, it is synsets — not word forms or lemmas — that participate in most WordNet relations.

The eight senses of the noun bass

I.3 Figure I.1

bass[1] the lowest part of the musical range

bass[2]**, bass part**[1] the lowest part in polyphonic music

bass[3]**, basso**[1] an adult male singer with the lowest voice

sea bass[1]**, bass**[4] the lean flesh of a saltwater fish (Serranidae)

freshwater bass[1]**, bass**[5] North American freshwater fish with lean flesh

bass[6]**, bass voice**[1]**, basso**[2] the lowest adult male singing voice

bass[7] the member with the lowest range of a family of instruments

bass[8] nontechnical name for edible marine and freshwater spiny-finned fishes

Each sense carries a gloss, a synonym list, and sometimes usage examples. WordNet has no pronunciation, so it cannot separate [b ae s] (bass[4], bass[5], bass[8]) from [b ey s] (the rest).

Synsets: WordNet's primitive unit

A synset is the set of near-synonyms that can express one concept. Instead of logical terms, WordNet represents a concept as a list of senses.

One synset, nine ways to say it

chump[1] | fool[2] | gull[1] | mark[9] | patsy[1] | fall guy[1] | sucker[1] | soft touch[1] | mug[2]

Gloss: *a person who is gullible and easy to take advantage of.*

Supersenses — 26 lexicographic categories for nouns (Fig. I.2)

ACT *service*	**ANIMAL** *dog*	**ARTIFACT** *car*	**ATTRIBUTE** *quality*	**BODY** *hair*	**COGNITION** *way*	**COMMUNICATION** *review*
FEELING *discomfort*	**FOOD** *food*	**GROUP** *place*	**LOCATION** *area*	**MOTIVE** *reason*	**NATURAL EVENT** *experience*	**NATURAL OBJECT** *flower*
OTHER *stuff*	**PERSON** *people*	**PHENOMENON** *result*	**PLANT** *tree*	**POSSESSION** *price*	**PROCESS** *process*	**QUANTITY** *amount*
RELATION *portion*	**SHAPE** *square*	**STATE** *pain*	**SUBSTANCE** *oil*	**TIME** *day*		

Also 15 for verbs, 2 for adjectives, 1 for adverbs. Supersenses are the fallback when WordNet senses are too fine-grained.

The WordNet relation inventory

Figures I.3 and I.4

Noun relations

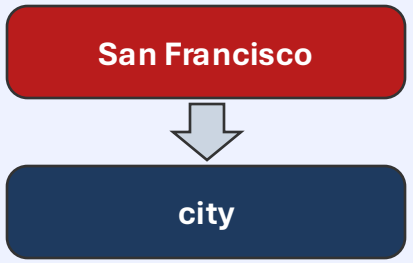
Relation	Definition	Example
Hypernym	Concept → superordinate	breakfast[1] → meal[1]
Hyponym	Concept → subtype	meal[1] → lunch[1]
Instance Hypernym	Instance → its concept	Austen[1] → author[1]
Instance Hyponym	Concept → its instances	composer[1] → Bach[1]
Part Meronym	Whole → part	table[2] → leg[3]
Part Holonym	Part → whole	course[7] → meal[1]
Antonym	Semantic opposition	leader[1] ⇄ follower[1]
Derivation	Same morphological root	destruction[1] ⇄ destroy[1]

Verb relations

Relation	Definition	Example
Hypernym	Event → superordinate event	fly[9] → travel[5]
Troponym	Event → subordinate event	walk[1] → stroll[1]
Entails	Event → event it entails	snore[1] → sleep[1]
Antonym	Semantic opposition	increase[1] ⇄ decrease[1]

Classes vs. instances

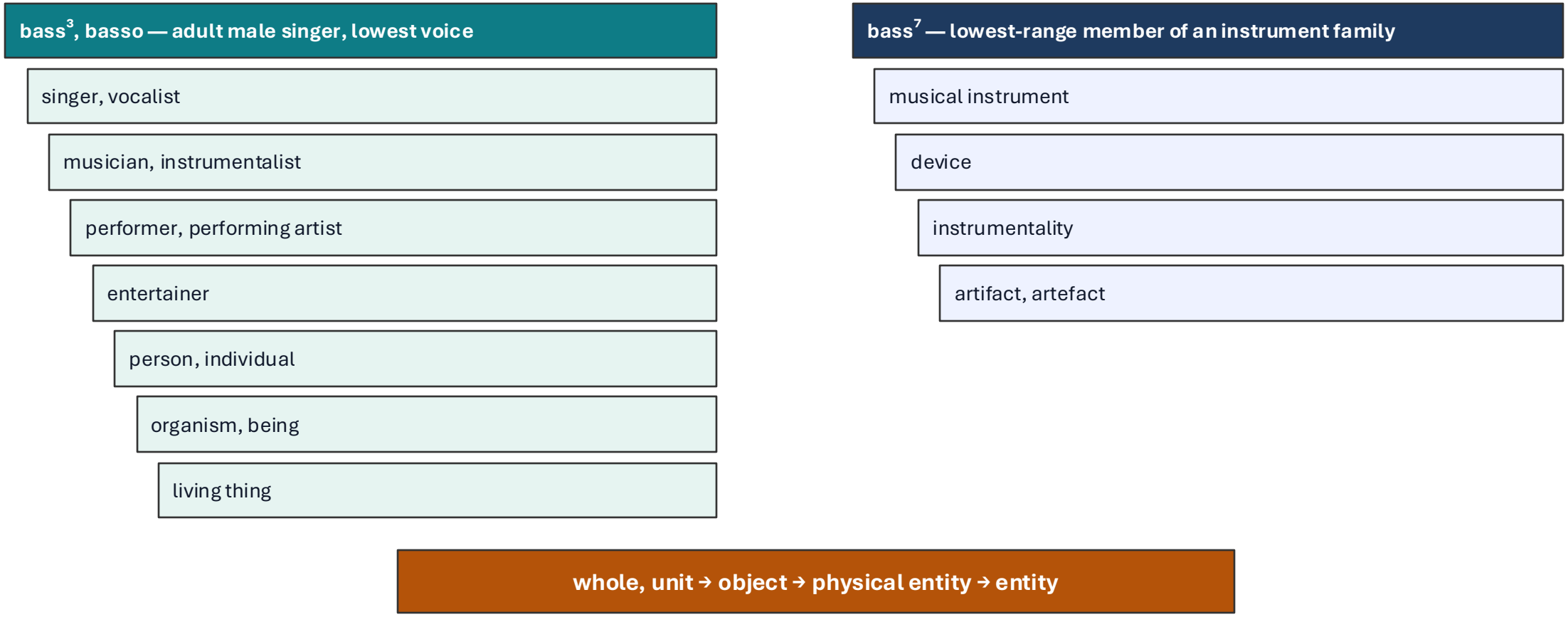
WordNet distinguishes two kinds of taxonomic entity.



Hypernym chains for bass[3] and bass[7]

I.3.1 Figure I.5

Following direct hypernym links produces a chain to ever more general synsets. The two chains are completely distinct, converging only at whole, unit.



04

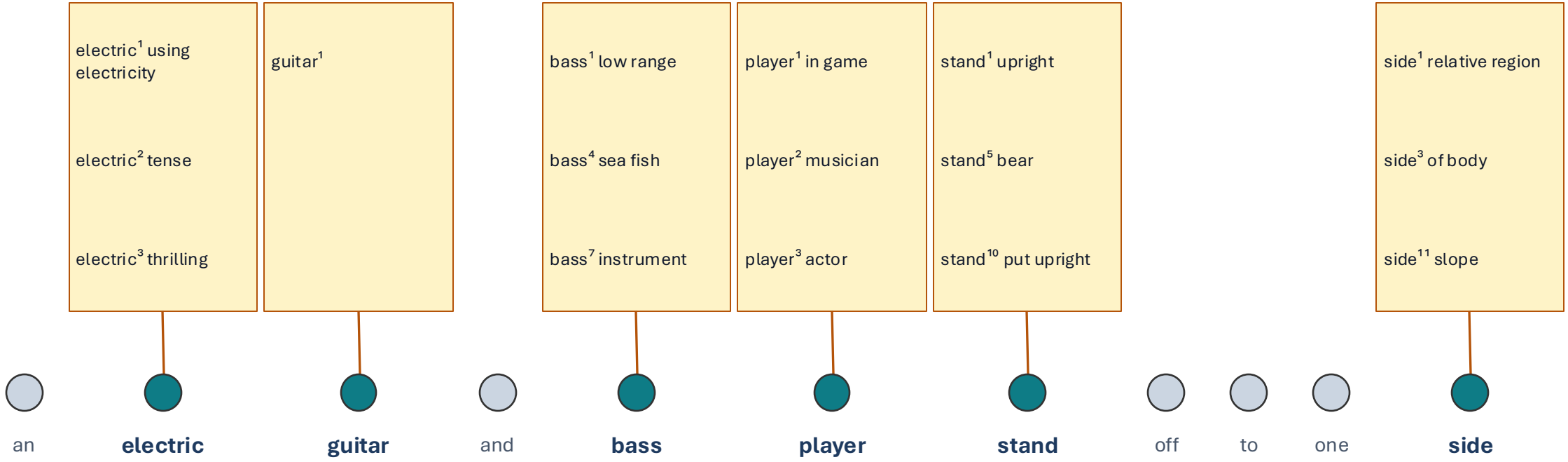
Word Sense Disambiguation

Given a word in context and a sense inventory, pick the right sense.

The all-words WSD task

Figure I.8

Input: a word in context plus a fixed inventory of senses. Output: the correct sense in that context. Only nouns, verbs, adjectives and adverbs are mapped.



Lexical sample task
A few pre-selected target words. Small label space — simple supervised classifiers work very well.

All-words task
Every content word in the text. Like POS tagging, but each lemma has its own tag set — so data is sparse.

Sense inventories and training data

I.4.1 Figure I.7

The inventory depends on the application: Spanish translations for MT, MeSH headings for medical indexing, WordNet senses or supersenses in general.

WordNet sense	Spanish translation	Supersense	Target word in context
bass^{4}	lubina	FOOD	... fish as Pacific salmon and striped bass and ...
bass^{7}	bajo	ARTIFACT	... play bass because he doesn't have to solo ...

Semantic concordance

A corpus where every open-class word is hand-labelled with a sense from a dictionary or thesaurus.

SemCor

A subset of the Brown Corpus — 226,036 words manually tagged with WordNet senses.

SENSEVAL / SemEval

SENSEVAL-3 English all-words: 2,282 annotations. Also SemEval-13 Task 12; Dutch and German corpora exist.

SemCor annotation — a subscript marks part of speech: **You will find$^{9}_{v}$ that avocado$^{1}_{n}$ is$^{1}_{v}$ unlike$^{1}_{j}$ other$^{1}_{j}$ fruit$^{1}_{n}$ you have ever$^{1}_{r}$ tasted$^{2}_{v}$**

Two strong baselines

Most frequent sense

Always pick the sense that is most frequent in a labelled corpus. In WordNet this is simply the first sense, because senses are ordered by their SemCor counts.

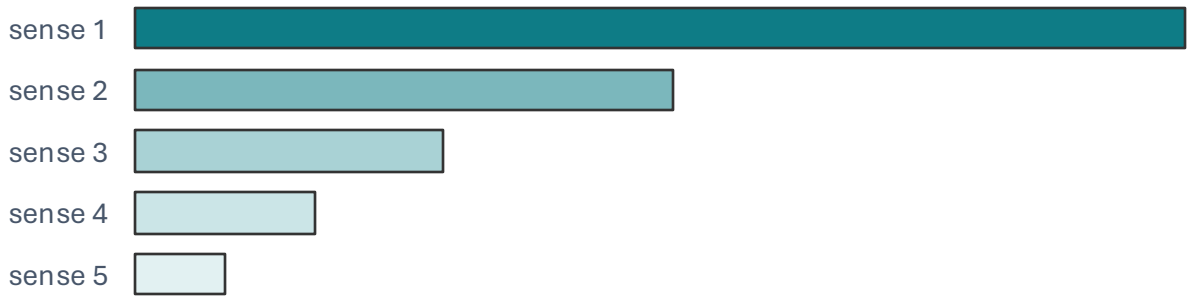
Surprisingly accurate — and the standard default when a supervised system has too little training data.

One sense per discourse

A word that appears several times in one text usually appears with the same sense (Gale et al., 1992b).

Holds best for coarse-grained senses and unrelated senses, so it is not used as a standalone baseline — but many systems keep a bias toward resolving an ambiguity consistently within a discourse.

Why “first sense” works: WordNet orders senses by corpus frequency



Frequency in SemCor falls off sharply after the first sense — so guessing sense 1 is right surprisingly often.

1-nearest-neighbour WSD

Melamud et al. (2016); Peters et al. (2018). Simple, and still the strongest approach.

Training

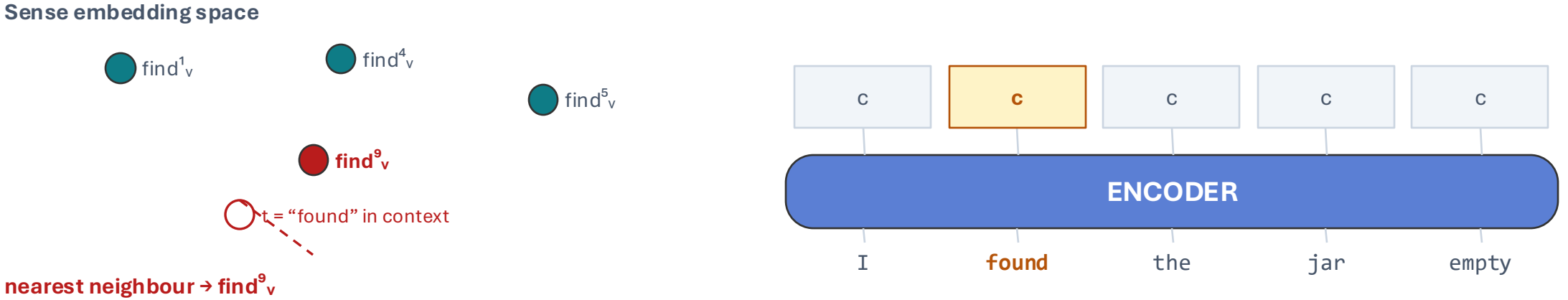
SemCor sentence → BERT encoder → token vectors v_i → average per sense

$$v_s = (1/n)\ \Sigma_i\ v_i \quad \text{for all } v_i \in \text{tokens}(s)$$

Test

Embed the target token t in context, then take the sense whose embedding has the highest cosine:

$$\text{sense}(t) = \text{argmax}_s\ \text{cosine}(t, v_s)$$



Senses never seen in training

Loureiro & Jorge, 2019

Only a small fraction of WordNet senses appear in SemCor. Rather than falling back to the most frequent sense, impute the missing embedding bottom-up through the taxonomy.

1 — Synset members

Average the sense embeddings of the other members of ŝ's synset.

$$\text{if } |S\hat{s}| > 0: \quad v_{\hat{s}} = (1/|S\hat{s}|) \; \Sigma \; v$$

2 — Hypernym synsets

Otherwise, average the hypernym-specific synset embeddings.

$$\text{if } |H\hat{s}| > 0: \quad v_{\hat{s}} = (1/|H\hat{s}|) \; \Sigma \; v$$

3 — Supersense

Otherwise, average all synset embeddings in the lexicographic category.

$$\text{if } |L\hat{s}| > 0: \quad v_{\hat{s}} = (1/|L\hat{s}|) \; \Sigma \; v$$

Every supersense has some labelled data in SemCor, so the cascade is guaranteed to produce a representation for every possible sense — coarse, but never missing.

Feature-based WSD: IMS

I.5.1 Zhong & Ng, 2010

An SVM over simple features of the surrounding words performs almost as well as contextual language model algorithms.

Part-of-speech tags

Window of 3 words each side, stopping at sentence boundaries

Collocations

Words or n-grams of length 1–3 at a particular position in a 3-word window

Weighted embeddings

Average of embeddings over a 10-word window, weighted exponentially by distance

Worked example — disambiguating bass with a 2-word window

“An electric guitar and bass player stand off to one side.”

guitar	and	**bass**	player	stand
w_{i-2}	w_{i-1}	target	w_{i+1}	w_{i+2}

```
[ guitar, NN, and, CC, player, NN, stand, VB, “guitar and”, “player stand”, g(E(guitar), E(and), E(player), E(stand)) ]
```

Lesk: a knowledge-based baseline

Knowledge-based methods need no labelled data — only WordNet. Lesk picks the sense whose gloss shares the most words with the target's neighbourhood.

```
SIMPLIFIED LESK (word, sentence)

best-sense ← most frequent sense for word
max-overlap ← 0
context ← set of words in sentence
for each sense in senses of word do
    signature ← words in gloss + examples
    overlap ← COMPUTEOVERLAP(signature, context)
    if overlap > max-overlap then
        max-overlap ← overlap
        best-sense ← sense
end
return (best-sense)
```

COMPUTEOVERLAP counts words shared between the two sets, ignoring stop words.

Extensions

Weight by IDF

Downweight frequent function words in the overlap count.

Embedding cosine

Best performing: replace word overlap with embedding similarity between gloss and context (Basile et al., 2014).

Neural Lesk

Use the definitions to compute sense embeddings used directly in place of SemCor-trained ones.

Knowledge-based methods matter wherever a thesaurus exists but a sense-labelled corpus does not.

Lesk at work: which bank?

Context: *"The bank can guarantee **deposits** will eventually cover future tuition costs because it invests in adjustable-rate **mortgage** securities."*

bank[1] — overlap = 2

Gloss: a financial institution that accepts deposits and channels the money into lending activities

Examples: *"he cashed a check at the bank", "that bank holds the mortgage on my home"*

deposits | mortgage

bank[2] — overlap = 0

Gloss: sloping land (especially the slope beside a body of water)

Examples: *"they pulled the canoe up on the bank", "he sat on the bank of the river"*

no non-stopword shares the context

Two non-stopwords overlap for bank[1] and none for bank[2] — so Lesk chooses bank[1], the financial institution.

Word-in-Context (WiC)

Given two sentences containing the same target word, decide whether the word is used in the same sense. WSD is contextualised similarity; WiC sits between it and context-free word similarity.

F	***bed***	*There's a lot of trash on the bed of the river*	— *I keep a glass of water next to my bed when I sleep*
F	***justify***	*Justify the margins*	— *The end justifies the means*
T	***air***	*Air pollution*	— *Open a window and let in some air*
T	***window***	*The expanded window will give us time to catch the thieves*	— *You have a two-hour window of clear weather*

Clustering first
WordNet senses are too fine-grained, so WiC merges first-degree neighbours in the semantic graph and same-supersense pairs.

Baseline algorithm
Embed the target word in both sentences with BERT, take the cosine, and threshold it on a dev set.

Wikipedia as a source of sense labels

When a Wikipedia article links a mention to a concept page, that link is effectively a sense annotation — free supervision at scale.

BAR (LAW)	*In 1834, Sumner was admitted to the [[bar (law)|bar]] at the age of twenty-three.*
BAR (MUSIC)	*It is danced in 3/4 time, with the couple turning approx. 180 degrees every [[bar (music)|bar]].*
BAR (ESTABLISHMENT)	*Jenga is a popular beer in the [[bar (establishment)|bar]]s of Thailand.*

The catch — and BabelNet

Wikipedia concepts must first be mapped onto whatever sense inventory the application uses. Automatic mappers compare the WordNet synset, gloss and related senses against the Wikipedia page title, outgoing links and categories, and take the greatest lexical overlap. The resulting mapping produced BabelNet (Navigli & Ponzetto, 2012).

Thesauruses improve embeddings

I.6 Figure I.12

Static embeddings confuse antonyms: expensive sits very close to cheap. Thesaurus relations can pull them apart.

Target	GloVe nearest neighbours — before	After counter-fitting
east	west, north, south	eastward, eastern, easterly
expensive	pricey, cheaper, costly	costly, pricy, overpriced
British	American, Australian, Britain	Brits, London, BBC

Retraining

Modify the embedding objective itself to incorporate synonymy, antonymy or supersenses — for word2vec (Yu & Dredze, 2014; Nguyen et al., 2016) or for contextual models (Levine et al., 2020).

Post-hoc mapping

Leave training alone; afterwards learn a second mapping that pushes thesaurus synonyms together and antonyms apart. Called retrofitting (Faruqui et al., 2015) or counter-fitting (Mrkšić et al., 2016).

Word sense induction

I.7 Schütze, 1992/1998

Sense-labelled corpora are expensive. In WSI the senses of a word are discovered automatically by clustering its contexts — so we never name them, we just number them.

Training — induce the senses

1. For each token w_i of w, compute a context vector c
2. Cluster the context vectors into k groups — each cluster is a sense
3. Take each cluster centroid s_j as the sense vector

Disambiguating a new token t

1. Compute a context vector c for t
2. Retrieve all sense vectors s_j for w
3. Assign t to the closest sense vector s_j

Agglomerative clustering

Start with every instance in its own cluster, repeatedly merge the two most similar, stop at k clusters or a goodness threshold.

Evaluation is unsolved

Extrinsic is best (e.g. diversifying search results). Intrinsic needs a mapping to gold senses — cluster-overlap metrics exist, but none is standard.

Applications, Limitations and Open Issues

- Applications: machine translation, information retrieval, question answering, text-to-speech, sentiment analysis, semantic search.
- Sense granularity: WordNet's fine distinctions are hard even for human annotators (inter-annotator agreement around 70-80%).
- Coverage: static inventories age quickly and miss domain-specific and multi-word senses.
- Data sparsity: most senses are rare, so supervised systems see few or no examples of them.
- Large language models capture much sense information implicitly, but explicit inventories remain useful for auditable, interpretable systems.
- Open directions: multilingual and cross-lingual WSD, few-shot sense modelling, dynamic sense inventories.

References and Further Reading

- Jurafsky, D. and Martin, J. H. Speech and Language Processing, 3rd edition - chapter on Word Senses and WordNet.
- Fellbaum, C. (ed.) WordNet: An Electronic Lexical Database. MIT Press, 1998.
- Miller, G. A. WordNet: A Lexical Database for English. Communications of the ACM, 1995.
- Navigli, R. Word Sense Disambiguation: A Survey. ACM Computing Surveys, 2009.
- Yarowsky, D. Unsupervised Word Sense Disambiguation Rivaling Supervised Methods. ACL, 1995.
- Princeton WordNet - wordnet.princeton.edu; NLTK WordNet interface - nltk.org.